MODULE 3 · PERCEPTION OF SOUND · N3 · ADDING SOURCES, DISTANCE & HEARING

Twice the Sound Is Not Twice the Decibels

Guided note · Dr. Byungmoon Cho · PHY 105 Physics of Sound

SYNOPSIS

Last class you measured sound levels with your phone. Today we find out what a decibel actually measures. The decibel scale counts **powers of ten**, so adding a second singer adds only a few decibels, and walking away from a speaker drops the level by a fixed amount each time you double the distance. We finish with the question every musician should ask: **how loud is too loud, and for how long?**

PERFORMANCE EXPECTATIONS

- **PE1** Place a sound on the decibel ladder: convert between intensity (W/m^2) and level (dB) using powers of ten.
- **PE2** Predict how the level changes when the number of identical sources changes (×2, ×10).
- **PE3** Predict how the level changes when your distance from a source doubles or grows ×10.
- **PE4** Estimate a safe listening time from a sound level using the 85 dB / 3 dB rule.

Instructions: work with a partner. Use your phone as a sound meter (**phyphox** → **Audio Amplitude**). Whenever you see **PREDICT**, write your prediction *before* you measure.

1 · The decibel ladder

YOU ALREADY KNOW THIS

Last class you ranked sounds by their readings on your phone. Write two of them here:
Quietest sound: _____ at _____ dB Loudest sound: _____ at _____ dB

Intensity I is the sound power passing through each square meter, in W/m^2. The faintest sound an average ear can detect has intensity $I_0 = 10^{-12}$ W/m^2. This is the **threshold of hearing**, and we call it 0 dB.

Level (dB)	Intensity (W/m^2)	Powers of ten above I_0	Example sound
0	10^{-12}	0	Threshold of hearing
10	10^{-11}	1	
20			
30			
40			
50			
60			
70			
80			
90			
100			
110			

Name: _____

Level (dB)	Intensity (W/m^2)	Powers of ten above I_0	Example sound
120	1		Threshold of pain
130			

KEY RELATIONSHIP

The count-the-tens rule: level in dB = 10 × (number of _____ above I_0)

$$\beta = 10 \log_{10}(I / I_0) \text{ with } I_0 = 10^{-12} \text{ W/m}^2$$

- A sound has intensity 10^{-6} W/m^2. Its level is _____
- Clapping read 30 dB above talking. The clapping was how many times more intense? _____
- Is a 0 dB sound silent? _____
- Can a sound be -10 dB? _____
- Why “deci”-bel? _____

2 · Is twice as many twice as loud?

PREDICT: one person humming reads *X* dB. What will two people read? The whole class?
Two people: _____ Whole class: _____

People humming	Measured level (dB)	Change from 1 person (dB)	Theory: 10 log(n) (dB)
1		0	0
2			____
4			____
8			____
Whole class, n = ____			____

WHAT WE FOUND

Doubling the intensity (×2) adds about ____ dB.

Ten times the intensity (×10) adds ____ dB.

A hundred times the intensity (×100) adds ____ dB.

Your ear hears +10 dB as roughly _____ as loud.

3 · The inverse-square walk

A small source spreads its sound power *P* over a growing sphere of area $4\pi r^2$:

$$I = P / (4\pi r^2)$$

PREDICT: if you double your distance *r*, the intensity becomes _____ as large, so the level drops by about ____ dB.

Distance from phone (m)	Measured level (dB)	Drop from previous (dB)	Theory: drop (dB)
0.5		—	—
1			___

Distance from phone (m)	Measured level (dB)	Drop from previous (dB)	Theory: drop (dB)
2			___
4			___

Explain: your measured drops were probably smaller than 6 dB. Why?

4 · Practice problems

Use the count-the-tens rule and the ×2 → +3 dB rule. Show your reasoning.

1. Conversation has intensity 10^{-6} W/m^2. What level is that in dB?

2. An 80 dB sound has what intensity?

3. One violin plays at 70 dB. What level do two violins make? Ten violins?

4. A speaker reads 100 dB at 2 m. What does it read at 4 m? At 20 m?

5. How many times more intense is 90 dB than 60 dB? About how many times louder does it sound?

5 · How loud is too loud? (Musicians' hearing)

Hearing damage depends on **level** and **time**. The recommended limit is 85 dB for 8 hours. Every additional 3 dB doubles the intensity, so it **halves** the safe time.

Level (dB)	85	88	91	94	97	100
Safe time	8 h	4 h	____	____	_____	_____

My earbuds at max volume read _____ dB on my phone, so they are safe for about _____ .

Caution: phone meters are not calibrated, so treat that number as a rough estimate. Differences between readings are trustworthy; the absolute number is not.

EXIT TICKET

Four identical speakers each produce 60 dB at your seat. What is the total level?

Total level: _____